

1. If $g(x) = x^3$ and $gf(x) = x^3 - 3x^2 + 3x - 1$, what is $f(x)$?
A) $x - 1$ B) $x + 1$ C) $x^2 + 1$ D) $x^2 - 1$
2. The sides of an equilateral triangle are shortened by 12 units, 13 units and 14 units respectively and a right angle triangle is formed. Then length of each side of the equilateral triangle is
A) 13 B) 15 C) 19 D) 17
3. The sum of the roots of the cubic equation $x^3 + 4x^2 + 5x + 2 = 0$ is equal to $-\alpha$. What is the value of α ?
A) 1 B) 4 C) 5 D) 3
4. Let the sequence $\{a_n\}$ be defined by $a_n = \frac{3n^2 - 5n + 2}{2n^2 + 7n - 1}$, then $\lim_{n \rightarrow \infty} a_n$ is
A) $\frac{5}{7}$ B) $\frac{2}{3}$ C) $\frac{3}{2}$ D) none of these
5. The equation of an ellipse is given by $\frac{x^2}{16} + \frac{y^2}{9} = 1$. If a tangent is drawn to the ellipse at the point $\left(2, \frac{3}{2}\right)$, what is the equation of the tangent line?
A) $2x + 3y = 5$ B) $4x + 3y = 11$ C) $4x + 6y = 11$ D) $3x + 4y = 24$
6. Value of the definite integral $\int_0^1 x(1-x)^{99} dx$ is equal to
A) $\frac{1}{11000}$ B) $\frac{1}{10010}$ C) $\frac{1}{10001}$ D) $\frac{1}{10100}$
7. A box contains 3 red balls, 4 green balls, and 5 blue balls. One ball is picked at random. What is the probability that the ball is not green?
A) $\frac{1}{2}$ B) $\frac{2}{3}$ C) $\frac{3}{4}$ D) $\frac{2}{5}$
8. $e^z = 1$ if and only if
A) $z = 0$ B) $z = 2\pi$ C) $z = 2\pi i$ D) $z = \text{an integral multiple of } 2\pi i$

9. If $z = x + iy$ is a complex number such that the real part of $\frac{1}{z}$ is equal to $\frac{1}{4}$, then the locus of z is
- A) A circle with center $(2, 0)$ and radius 2
 B) A circle with center $(0, 2)$ and radius 2
 C) A circle with center $(-2, 0)$ and radius 2
 D) A straight line passing through the origin.
10. If the function $f(x)$ is defined by the rule
- $$f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ ax + b & \text{if } x > 1 \end{cases}$$
- then find the values of a and b for which the function is differentiable at $x = 1$
- A) $a = 2$ and $b = -1$ B) $a = -1$ and $b = 2$
 C) $a = 1$ and $b = 0$ D) $a = 2$ and $b = 1$
11. If F is an open covering of a closed and bounded set A in E_n , the n dimensional Euclidean space, then Heine - Borel theorem states that
- A) a finite sub collection of F also covers A
 B) a countable sub collection of F also covers A
 C) any sub collection of F also covers A
 D) None of these
12. The series $\sum_{n=1}^{\infty} n^{1-n}$ is
- A) convergent
 B) divergent
 C) neither convergent nor divergent
 D) none of these
13. If $\mathbb{Q}$ is the set of all rational numbers, then
- A) $\mathbb{Q}$ is finite B) $\mathbb{Q}$ countable C) $\mathbb{Q}$ uncountable D) None of these
14. What is the limit of the sequence $\{a_n\}$ defined by $a_n = \frac{e^n}{n!}$ as n approaches infinity?
- A) e B) 1 C) 0 D) the sequence diverges to ∞
15. Polar form of the Cauchy Riemann equations is
- A) $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$ B) $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$
 C) $\frac{\partial u}{\partial r} = -\frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = \frac{1}{r} \frac{\partial u}{\partial \theta}$ D) $\frac{\partial u}{\partial r} = r \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -r \frac{\partial u}{\partial \theta}$

16. Let $f(x) = e^x \cos x$. The third-degree Taylor polynomial of $f(x)$ about $x = 0$ is:
- A) $1 + x + \frac{x^2}{2} + \frac{x^3}{3}$ B) $1 + x - \frac{x^2}{2} - \frac{x^3}{3}$
 C) $1 + x - \frac{x^2}{2} + \frac{x^3}{3}$ D) $1 + x - \frac{x^3}{3}$
17. Consider the function $f(z) = \frac{e^z}{(z-1)^2}$. What is the residue of $f(z)$ at its singular point $z_0 = 1$?
- A) 0 B) 1 C) e D) e^2
18. Value of the integral $\int_C \frac{z^{100}}{z^{101} + 1} dZ$ where C is the circle centred at the origin radius 2 units and taken in the anti clockwise direction is
- A) $-2\pi i$ B) 2π C) 0 D) $2\pi i$
19. Consider the statements
- i) In a group the two cancellation laws hold
 ii) Closed finite subset of a group is a group
 iii) Set of automorphisms of a group is a group
- A) (i) and (ii) are true but (iii) is not true
 B) (ii) and (iii) are true but (i) is not true
 C) (iii) and (i) are true but (ii) is not true
 D) all the three statements are true
20. If G is a group with identity e and G' is the commutator subgroup of G , then G is abelian if and only if G' is equal to
- A) G B) $\{e\}$ C) empty set D) none of these
21. The set of solutions of the equation $x^3 - 2x^2 - 3x = 0$ in $\mathbb{Z}_{12}$ is
- A) $\{0, 3, 5, 8, 9, 11\}$ B) $\{0, 2, 4, 6, 8, 10\}$ C) $\{0, 3, 5, 8, 9, 10\}$ D) $\{0, 3, 5, 7, 9, 11\}$
22. Consider the statements
- i. The field of rational numbers $\mathbb{Q}$ is a field of quotients of the $\mathbb{Z}$, the ring of integers.
 ii. The field of real numbers $\mathbb{R}$ is a field of quotients of $\mathbb{R}$.
- Which among the following is correct?
- A) (i) is correct but (ii) is wrong B) (i) is wrong but (ii) is correct
 C) both (i) and (ii) are wrong D) both (i) and (ii) are correct
23. Consider the statements
- i. If $\mathbb{R}$ is a commutative ring, then $\mathbb{R}[x]$ is commutative.
 ii. If $\mathbb{D}$ is an integral domain, then $\mathbb{D}[x]$ is an integral domain.
- Which among the following is correct?
- A) (i) is correct but (ii) is wrong B) (i) is wrong but (ii) is correct
 C) both (i) and (ii) are wrong D) both (i) and (ii) are correct

24. A Sylow 3-subgroup of a group of order 54 has order
 A) 54 B) 38 C) 27 D) 18
25. The matrix $A = \begin{pmatrix} 3 & 1 & x \\ 4 & 2 & 5 \\ 1 & 0 & 1 \end{pmatrix}$ is singular if
 A) $x = 1$ B) $x = \frac{7}{2}$ C) $x = \frac{1}{4}$ D) $x = 0$
26. Let A be a 3×3 matrix such that $\det(A) = 4$. If $B = 2A^T$, what is the value of $\det(B)$?
 A) 4 B) 16 C) 32 D) none of these
27. Let V be a vector space with $\dim(V) = n$, where n is a positive integer. Consider a set of m vectors $S = \{v_1, v_2, \dots, v_m\}$ in V . Which of the following statement is always true?
 A) If $m < n$, then the set S must span V .
 B) If $m > n$, then the set S must be linearly independent.
 C) If $m = n$ and S spans V , then S must also be linearly independent.
 D) If $m = n$, then S must be a basis for V .
28. The product $\gcd(197, 48) \times \text{lcm}(197, 48)$ is equal to
 A) 197 B) 48 C) 9456 D) 9654
29. Which of the following is a basis for M_{22} , the vector space of all 2×2 real matrices?
 A) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. B) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$.
 C) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. D) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$.
30. Let $T : \mathbb{R}^9 \rightarrow \mathbb{R}^9$ be a linear transformation such that $\dim(\ker T) = 2$. Which of the following statements is true?
 A) T is invertible B) T is one-to-one but not onto
 C) T is onto but not one-to-one D) T is neither one-to-one nor onto
31. For the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(a, b) = (a - b, b - a, -a)$, what is the rank of T ?
 A) 0 B) 1 C) 2 D) 3
32. If p is a prime number, which of the following expressions correctly gives the formula for $\phi(p^k)$, where k is a positive integer and ϕ is the Euler's totient function ?
 A) $p^k - p^{k-1}$ B) $p^k - 1$ C) $p^{k+1} - p^k$ D) p^{k-1}
33. If r is the remainder when $225!$ (! denotes factorial) is divided by 227, then
 A) $r = 0$ B) $r = 1$ C) $r = 2$ D) $r = 226$

34. By eliminating the arbitrary constants a and b from the equation $y = e^{3x}(a \cos 2x + b \sin 2x)$, we get the differential equation

A) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = 0$

B) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 5y = 0$

C) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 4y = 0$

D) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$

35. The one dimensional wave equation

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2} \text{ is}$$

A) parabolic

B) elliptic

C) hyperbolic

D) none of these

36. The complete integral of the partial differential equation $q = 3p^2$ is

A) $z = ax + 3a^2y + b$

B) $z = ax - 3a^2y + b$

C) $z = -ax + 3ay + b$

D) none of these

37. In a metric space (X, d) , $B(a, r)$ denotes the open ball centered at a and radius r . Consider the statements:

i. $B(a, r)$ is contained in a closed ball centered at a

ii. $B(a, r)$ contains a closed ball centered at a

Then which is the correct choice?

A) (i) is true and (ii) is false

B) (i) is false and (ii) is true

C) both (i) and (ii) are true

D) both (i) and (ii) are false

38. Let X be an infinite set and T_f be the collection of all subsets U of X such that either $X - U$ is finite or all of X . Similarly T_c is the collection of all subsets V of X such that either $X - V$ is countable or all of X .

Which of the following statements is true?

A) T_c and T_f are topologies on X and T_c is finer than T_f

B) T_c and T_f are topologies on X and T_f is finer than T_c

C) T_c and T_f are topologies on X and $T_c = T_f$

D) T_c and T_f are not topologies on X

39. Which among the following is a "false" statement?
- A) Metrizable is a hereditary property
 - B) Second countability is a hereditary property
 - C) Compactness is a hereditary property
 - D) Connectedness is not a hereditary property
40. Consider the two statements
- i. Every separable topological space is second countable
 - ii. Every second countable topological space is separable
- Which of the following is the correct option?
- A) both (i) and (ii) are true
 - B) (i) is true and (ii) is false
 - C) (i) is false and (ii) is true
 - D) both (i) and (ii) are false
41. Let H be a Hilbert space, and let $M \subset H$ be a closed subspace. For any vector $x \in H$, which of the following statements is always true?
- A) There exists a unique vector $y \in M$ such that $x - y$ is orthogonal to every vector in M
 - B) x always belongs to M
 - C) The distance from x to M may not be attained
 - D) The projection of x onto M may not be unique
42. Let X be a Banach space. Which of the following statements is always true?
- A) If X is separable, then X^* is also separable
 - B) If X^* is separable, then X is separable
 - C) Every bounded linear functional on X^* is given by evaluation at some $x \in X$
 - D) The closed unit ball in X^* is compact in the norm topology
43. If E is an orthonormal subset of an inner product space X , then for $u, v \in E$ with $u \neq v$, the value of $\|u - v\|$ is
- A) 2
 - B) $\sqrt{2}$
 - C) 2^p
 - D) $2^{\frac{1}{p}}$

44. The domain of the function $f(x) = \sqrt{x} + \sqrt{2x-1}$ is
 A) $[0, \infty)$ B) $[\frac{1}{2}, \infty)$
 C) $\mathbb{R}$, the set of real numbers D) $[1, \infty)$
45. Which of the following is 'not' true?
 A) $A \cap B = \emptyset \Leftrightarrow A \cap \bar{B} = A$
 B) $A \cap B \neq \emptyset \Rightarrow A \neq \emptyset$ and $B \neq \emptyset$
 C) $A \cap B = \emptyset \Rightarrow A = \emptyset$ or $B = \emptyset$
 D) $A \neq B \Rightarrow \bar{A} \neq \bar{B}$
46. The equation of the plane passing through the intersection of the planes $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$ and the point $(1,1,1)$ is
 A) $21x + 19y + 23z = 45$ B) $2x + 15y + 3z = 6$
 C) $13x - 24y + 12z = 21$ D) $20x + 23y + 26z = 69$
47. The equation of the sphere described on the join of the points $(2, -3, 4)$ and $(-5, 6, -7)$ as diameter.
 A) $x^2 + y^2 + z^2 + 3x - 3y - 3z - 65 = 0$
 B) $x^2 + y^2 + z^2 + 2x + 3y - z - 5 = 0$
 C) $x^2 + y^2 + z^2 - 3x - 3y - z - 24 = 0$
 D) $x^2 + y^2 + z^2 + 3x - 3y + 3z - 56 = 0$
48. If $x = at^2$ and $y = 2at$, then $\frac{dy}{dx} =$
 A) 1 B) $\frac{1}{t}$ C) t^2 D) $\frac{1}{t^2}$
49. The interval in which the function $f(x) = 6 - 9x - x^2$ is increasing
 A) $[2, 9]$ B) $(-\infty, \frac{9}{2}]$ C) $(-\infty, -\frac{9}{2}]$ D) $[-\frac{9}{2}, \infty)$
50. Let X be a continuous random variable with density $f(x) = kx^2$, $0 \leq x \leq 2$. Then $P(1 \leq x \leq 2)$
 A) $\frac{2}{7}$ B) $\frac{7}{8}$ C) $\frac{2}{3}$ D) $\frac{5}{8}$

51. Which of the following is 'not' uniformly continuous?
- A) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$
 - B) $f: [0,1] \rightarrow \mathbb{R}$ defined by $f(x) = x^3$
 - C) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 5x$
 - D) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$
52. Evaluate $\lim_{n \rightarrow \infty} \sum_{r=0}^{n-1} \frac{n}{n^2+r^2}$
- A) 0
 - B) 11
 - C) $\frac{\pi}{2}$
 - D) $\frac{\pi}{4}$
53. Which of the following is 'not' true?
- A) If $f: [a, b] \rightarrow \mathbb{R}$ is monotone, then f is a function of bounded variation on $[a, b]$
 - B) If $f: [a, b] \rightarrow \mathbb{R}$ is continuous, then f is a function of bounded variation on $[a, b]$
 - C) $f(x) = x^{1/3}$ is of bounded variation on every finite interval.
 - D) If f is a function of bounded variation on $[a, b]$ then f is bounded on $[a, b]$
54. The function $f(z) = \frac{1}{\sin(\frac{1}{z})}$ has
- A) a simple pole at $z = 0$
 - B) a removable singularity at $z = 0$
 - C) a non-isolated essential singularity at $z = 0$
 - D) an isolated essential singularity at $z = 0$
55. The gcd of 2 and $-1 + 5i$ in $\mathbb{Z}[i]$, the ring of Gaussian integers is:
- A) 1
 - B) 2
 - C) $-1 + 5i$
 - D) $1 - i$
56. Which of the following is 'not' true?
- A) The set of all rational numbers x such that $0 \leq x \leq 1$ is a group under ordinary multiplication
 - B) The set of all positive rational numbers forms a group under the binary operation $*$ defined by $a * b = \frac{ab}{2}$
 - C) The set Q^* of all non-zero rational numbers forms a group under ordinary multiplication.
 - D) The set of all rational numbers of the form $2^a 3^b$, a, b are integers forms a group under ordinary multiplication.

57. Which of the following is 'not' true?
- A) Any subgroup of a solvable group is solvable
 - B) A group of prime power order is solvable
 - C) Any abelian group is solvable
 - D) S_5 , the permutation group on 5 symbols is solvable
58. Which of the following is 'not' true?
- A) If F is a field, then $F[x]$ is a Euclidean domain
 - B) If F is a field, then $F[x]$ is a field
 - C) If F is a field, then $F[x]$ is a principal ideal domain
 - D) If F is a field, then $F[x]$ is a unique factorisation domain
59. Let $\varphi: \mathbb{Z}[x] \rightarrow \mathbb{Z}$ given by $\varphi(f(x)) = f(0)$ be an onto mapping. Then which of the following is 'not' true?
- A) φ is a ring homomorphism with $\ker(\varphi) = \langle x \rangle$
 - B) $\frac{\mathbb{Z}[x]}{\langle x \rangle} \approx \mathbb{Z}$
 - C) The ideal $\langle x \rangle$ is prime
 - D) The ideal $\langle x \rangle$ is maximal
60. Which of the following is 'not' true about the polynomial $f(x) = x^4 + 4$
- A) $f(x)$ is reducible over $\mathbb{Z}$
 - B) $f(x)$ is irreducible over $\mathbb{Q}$
 - C) $f(x)$ is reducible over $\mathbb{R}$
 - D) $f(x)$ is reducible over $\mathbb{C}$
61. The splitting field of the polynomial $x^3 - 2$ over the field $\mathbb{Q}$ is
- A) $\mathbb{Q}(\omega)$
 - B) $\mathbb{Q}(\sqrt{2})$
 - C) $\mathbb{Q}(\sqrt[3]{2})$
 - D) $\mathbb{Q}(\omega, \sqrt[3]{2})$
62. If the nullity of the matrix $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & k \end{pmatrix}$ is 1, then the value of k is
- A) -1
 - B) 0
 - C) 1
 - D) 2
63. Consider the system of linear equations
- $$\begin{aligned} x + y + z &= 1 \\ 2x - y + 3z &= 4 \\ 3x + 2y + kz &= 0 \end{aligned}$$
- Then the value of k for which the system has no solution
- A) $k = -1$
 - B) $k = -5$
 - C) $k = \frac{10}{3}$
 - D) $k = 7$

64. A matrix with characteristic polynomial $(x - 1)^3(x - 2)$ is diagonalizable if and only if
 A) Rank of the matrix is 4
 B) Algebraic multiplicity and geometric multiplicity of the eigen value 1 are same
 C) Trace of the matrix is 5
 D) Determinant of the matrix is 2
65. Which of the following is 'not' true about similar matrices A and B of order n
 A) There exists a non-singular matrix S such that $S^{-1}AS = B$
 B) A and B have the same characteristic polynomials
 C) A and B have the same eigen values
 D) A and B have the same eigen vectors
66. The dimension of the subspace $W = \{(x, y, z) \in \mathbb{R}^3 | x - y + z = 0, x - y = 0\}$ of $\mathbb{R}^3$ is
 A) 0 B) 1 C) 2 D) 3
67. The last digit of the number $(2026)^{2026}$
 A) 2 B) 4 C) 6 D) 8
68. Which of the following is a linear transformation?
 A) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(x, y) = (x, xy)$
 B) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by $T(x, y, z) = (2x + y, x + z)$
 C) $T: \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = 4x + 3$
 D) $T: \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = \cos x$
69. The number of solutions of the congruence $x^2 \equiv 1 \pmod{35}$ is
 A) 2 B) 4 C) 6 D) 8
70. The order of the differential equation $\left(\frac{d^2y}{dx^2}\right)^4 + \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = 0$ are
 A) order = 1 B) order = 2 C) order = 4 D) order = 8
71. The integrating factor of $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$
 A) x^{-2} B) y^{-2} C) $x^{-2}y^{-2}$ D) None of these
72. The solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x-y}$ is
 A) $x - y = c$ B) $x^2 - y^2 = c$
 C) $\tan^{-1}\left(\frac{y}{x}\right) = c$ D) None of these

73. The orthogonal trajectories of the family of curves $y = ce^x$ is
 A) $y = ce^{-x}$ B) $x^2 + y^2 = c$
 C) $2x + y^2 = c$ D) $y = cx^2$
74. The solution of the partial differential equation $\frac{\partial^2 z}{\partial x^2} - z = 0$ given when $x = 0, z = e^{2y}$ and $\frac{\partial z}{\partial x} = 3e^{2y}$
 A) $z = e^{2y}(\cosh x + 3\sinh x)$ B) $z = e^{2y}(\cosh x - 3\sinh x)$
 C) $z = e^{2y}(\sinh x + 3\cosh x)$ D) $z = e^{2y}(\sinh x - 3\cosh x)$
75. Which of the following metric space is 'not' complete?
 A) $\mathbb{R}$ with usual metric
 B) $\mathbb{R}$ with discrete metric
 C) Open interval $(0,1)$ with usual metric
 D) $\mathbb{R}^n$ with metric $d(x, y) = ||x - y||$
76. Let $f: (X, d_1) \rightarrow (Y, d_2)$ be a continuous function. Then which of the following statements is 'not' true?
 A) Inverse image of every closed subset of Y is closed in X
 B) Image of every open subset of Y is open in X
 C) If f is bijective then f^{-1} is continuous
 D) For any subset A of $X, f(\overline{A}) \subseteq \overline{f(A)}$
77. Let X be an infinite set with cofinite topology. Then which of the following is true?
 A) X is T_0 , but not T_1 B) X is T_1 , but not T_0
 C) X is both T_0 and T_1 D) X is neither T_0 nor T_1
78. Which of the following normed space is 'not' separable?
 A) l_2 B) l_∞ C) $\mathbb{R}^n$ D) $C([0,1])$
79. Which of the following is 'not' a Hilbert Space?
 A) l_2
 B) $L_2[a, b]$, the square integrable functions
 C) $\mathbb{R}^n$ with $\langle x, y \rangle = \sum_{k=1}^n x_k y_k$
 D) $C([a, b])$, the space of continuous functions with $\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt$
80. Dual space of $c_0 = \{x = (x_n): (x_n) \rightarrow 0 \text{ as } n \rightarrow \infty\}$ is
 A) l_1 B) l_2 C) l_∞ D) None of these